# The National Hospital for Neurology and Neurosurgery
## Neuroscience Critical Care Unit (NCCU)

University College London Hospitals NHS
NHS Foundation Trust

# LIQUOGUARD AUTOMATED INTRACRANIAL PRESSURE CONTROLLED CSF DRAINAGE

1. **The drain is set up by the neurosurgeon**
   *Ensure* ***red bung*** *replaces vented bung*

**2.Keep plugged in and do not switch LiquoGuard machine off when in use at bedside**
- If LiquoGuard loses power or is switched off, it will return to factory settings and for some patients they may drain too much CSF – *machine will function on battery power for transferring patient*

**3.Zero reference point**
- As with all external ventricular drainage systems **zero reference point** is **tragus**
- With LIquoGuard system place **ECG dot** at temple and attach sensor to set zero reference point

**4.Settings**
- Drainage mL/h and pressure mmHg
- Will only drain when actual pressure (**white curve**) is higher than target pressure (**blue line**) – therefore it is possible CSF drainage will be less than set flow rate

**6.Monitor drainage volume** (record hourly)
- Select **Info** in bottom tabs and then select **volumes** from right hand menu
- The **pSet** (Pressure set) and **vSet** (Volume set) should be checked and documented every hour along with the **volume drained**

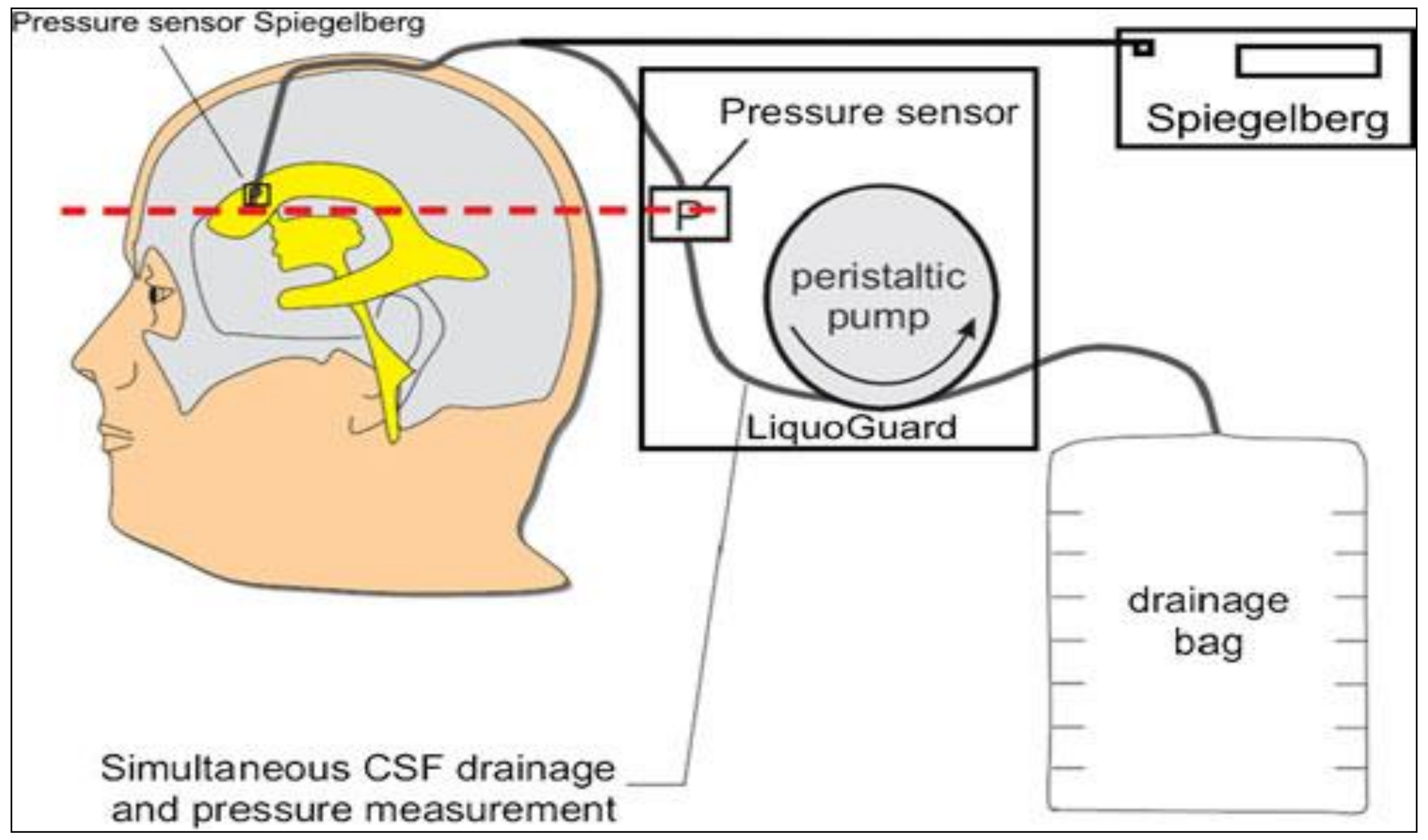
**5.Alarms -** If alarm goes off
- Silence / Troubleshoot / Seek help
- Pulsation alarms can be disabled when used as a **lumbar drain** as the waveform is flatter and the LiquoGuard machine may keep alarming

| | |
|---|---|
| **High/Low** | May trigger if patient sits up too rapidly – raise head of bed slowly |
| **Pulse** | Check if blocked or kinked |
| **Flow** | Too much / too little drainage |

**7.Taking patient to CT scan**
- Place in pause mode or keep running dependent on individual patient

**8.Raising or lowering height of bed / sitting patient out of bed**
- No need to pause –drainage is not dependent on relation of machine to zero reference point

**9.Special considerations**
- If pump alarms to say drainage set has expired *(manufacturer's advice change every 7 days)* but neurosurgeon happy to keep same set attached (usually up to 10 days), try changing LiquoGuard pump but keep same tubing attached
- Empty drainage bag when full - no need to change entire set
- **For MRI**, **drainage set is MRI compatible, LIquoGuard machine and ECG electrode** (used as sensor) **are not -** Drainage set needs to be removed by medical staff from LiquoGuard pump – replace after MRI without need to disconnect drain - *not required for CT / xray*



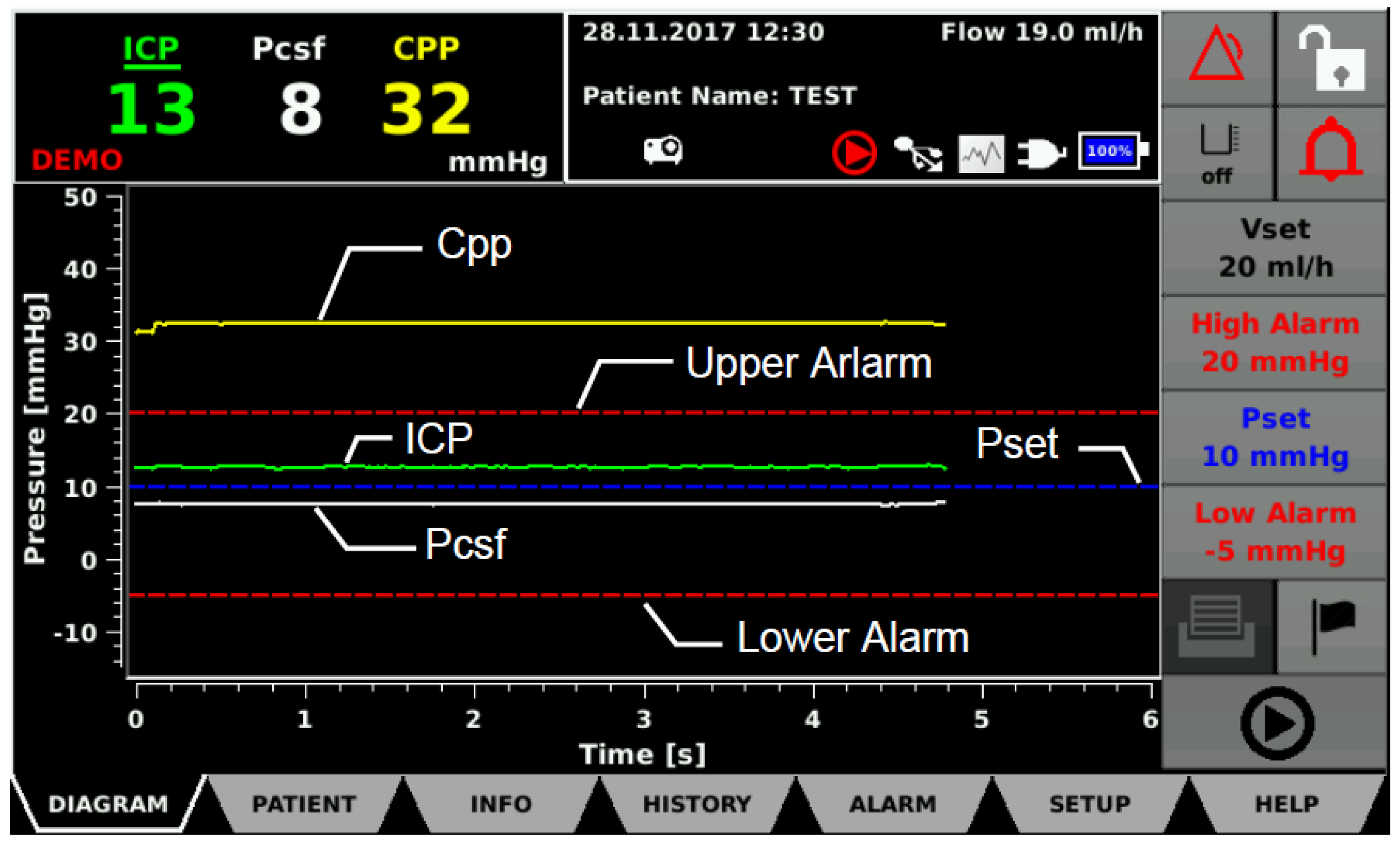




Figure 11: Limit values and curves shown in the diagram.

1. Pressure (y-axis) in mmHg or cmH2O
2. White curve: CSF pressure measured by pressure sensor in tubing set
3.
4. Green curve: ICP pressure value measured by an external sensor (parenchymal or catheter tip = ventricular)
5. Yellow curve: cerebral perfusion pressure (CPP)
6. Blue dotted line: *Pset* When CSF pressure (ICP or Pcsf) higher than Pset, and application not paused, pump turns (drainage) to reduce CSF pressure to Pset
7. Top red dotted line: *Upper Alarm* (upper pressure alarm limit)
8. Lower red dotted line: *Lower Alarm* (lower pressure alarm limit

| CSF SAMPLING |
|---|
| LiquoGuard set does not have a drip chamber therefore CSF samples must be taken from the 3-way tap – *follow CSF sampling protocol* |
| ADMINISTERING IT ANTIBIOTICS |
| LiquoGuard machine must be set to PAUSE MODE when administering IT antibiotics – *follow IT protocol* |